UNIVERSITY OF CALOOCAN CITY
COMPUTER ENGINEERING DEPARTMENT

Software Design

Progress Report No. 8

Data Structures and Algorithms

Submitted by:
Asugas, Kenneth R.
Delinia, Filjohn B.
Eulin, Ryan Bertrand B.
Hermosura, Leigh B.
Maringal, Czer Justine D.
Polestico, Paul Justine D.

Instructor:
Engr. Maria Rizette H. Sayo

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I. Objectives

In this section, the goals in this laboratory are:

- To describe a good conceptual design of the system
- To identify one of the three levels of structure of program – Data Structure

II. Methods

One of the most difficult things for program readers to understand is the way in which data is structured and used. A data map is very useful in interpreting the code's actions, especially when the system handles many files of varying types and purposes.

Why do we need Data Structure?

- It gives different level of organizing data
- It explains how data can be stored and accessed
- It provides operation on group of data, such as adding an item, inserting an item, sorting items, deleting an item, and updating items
- It provides a mean to manage huge amount of data efficiently
- It provides fast searching and sorting a data

III. Results

General Instructions:

Create an Algorithm and Flowchart, Data Flow Diagram/ERD of your proposed system (Use LucidChart to draw your flowchart and other drawing software)

ALGORITHM:

Phase 0: Security (Access Control)

- Step 1: Entry – User opens the website to Login or Register.
- Step 2: Authentication – System verifies the username and password.
- Step 3: Authorization – If correct, the user is granted access to the Dashboard; otherwise, access is denied.

Phase 1: Stock-In (Adding Items)

- Step 1: Input – Admin enters item details (type, size, and quantity).
- Step 2: Check – System checks if the item already exists in the database.
- If Yes: Update the existing total.
- If No: Create a new unique Item ID.
- Step 3: Log – Record the "PRODUCTION_IN" with the user's name and time.

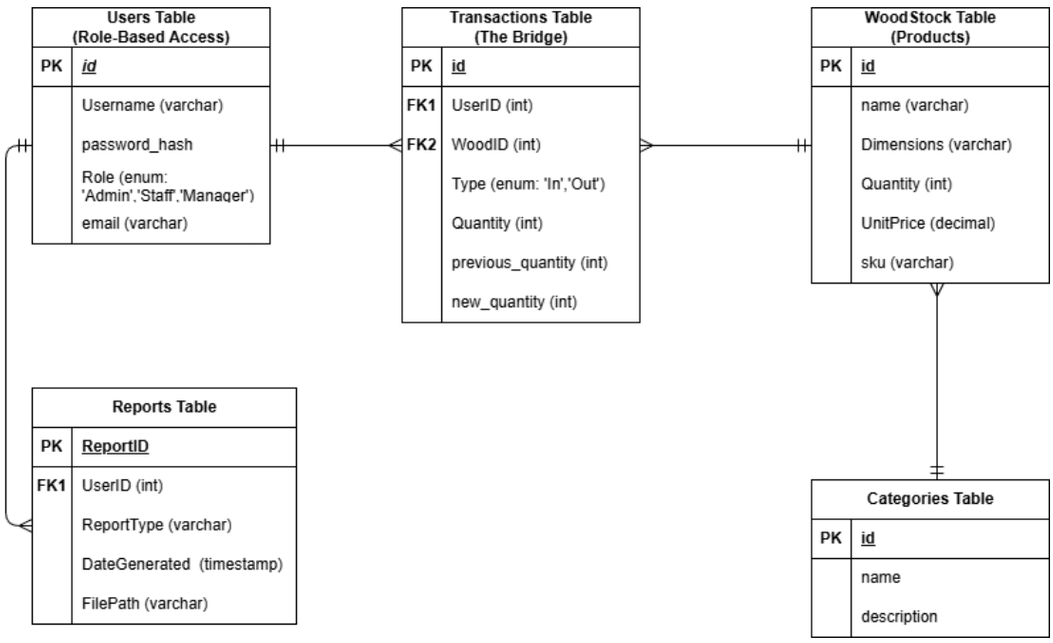
Phase 2: Stock-Out (Releasing Items)

- Step 1: Request – User selects an item and amount to release.
- Step 2: Guard Clause – System checks: Is Request > Current Stock?
- If Yes: Show "Insufficient Stock Error."
- If No: Subtract the amount from the database.
- Step 3: Log – Record the "OUT_OF_STOCK" for the transaction history.

Phase 3: Reporting (Hard Copy)

- Step 1: Request – User clicks "Export File."
- Step 2: Conversion – System pulls data and converts it into an Excel format.
- Step 3: Download – The file is saved for printing or archiving.

Entity Relationship Diagram (ERD) of the AWIC System



IV. Conclusion

This laboratory successfully identified the Data Structure as the foundational level of the AWIC System. By moving away from a "flat-file" structure (Excel), we established a Relational Data Model that ensures data integrity and efficiency.

By defining this structure, this program can efficiently execute algorithms for stock-in and stock-out while maintaining a "single source of truth" for Asia Wood International Corporation’s inventory.

References

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